

Effects of Endurance-, Strength-, and Concurrent Training on Cytokines and Inflammation

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Introduction

The immune system maintains physiological homeostasis and health by protecting the body from the harmful effects of antigens (molecules that induce an immune response), including microbes, pathogens, stress-related insults, and soft tissue damage. This system monitors antigenic insults using a two-tier response against antigens of varying complexity and novelty [1].

The first line of defence against foreign bodies involves the innate or non-adaptive immune system, which responds to foreign particles in the body using a generic mechanism. The protective mechanisms under this immune system include surface barriers (e.g., skin and mucosa surface), localized inflammation, phagocytosis by monocytes and granulocytes, and cytotoxicity by binding natural killer (NK) cells to the major histocompatibility complexes (MHC) of tumour or virus-infected cells. The second line of defence, known as the adaptive immune system, is activated against antigens that are not neutralized by the innate immune system. The immune cells in this system can recognize unique features in the antigen and modify its own morphology and functions to form new antibodies that target and neutralize specific novel antigens. These characteristics of the adaptive immune system allow

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the body to mount an antibody-response against novel antigens, and the new antibodies become a permanent component of the immune surveillance system in the body. This system is activated about 24–48 h after an antigenic insult and is driven by lymphocyte subset cells, i.e. T cells (cytotoxic T and T-helper cells) as well as B cells and immunoglobulin (Ig)G, IgA, IgM, and IgE [2].

Besides antigenic insults, the state of the immune system is also influenced by acute and prolonged physical stressors, such as endurance-, strength-, and sports training and occupational tasks (e.g., military and physical labour) [3–6]. Prolonged exposure to these stressors can lead to suppression of immune functions and skeletal muscle damage that triggers the inflammatory response. The inflammatory response is part of the tissue protection and repair functions in the immune system and is driven primarily by cytokines and chemokines. Cytokines and chemokines are intercellular signalling molecules that modulate inflammation and immune responses, usually in an autocrine or paracrine manner at extremely low concentrations [7]. In the context of exercise, the inflammatory response can involve intricate cellular signalling between different immune and non-immune cells (e.g. skeletal muscle), leading to the secretion and uptake of cytokines and chemokines and the downstream mobilization of leukocytes into the systemic circulation and inflammatory sites in the body. These inflammatory responses can also activate tissue resident immune cells and change their functional responses. Cytokines are generally classified as being pro- or anti-inflammatory, based on their effects on cellular or immune pathways and the plasma concentrations of anti-inflammatory cytokines (e.g. IL-1ra, IL-4, IL-10) and chemokines (e.g. IL-8) are typically upregulated after a bout of exercise ([7, 8]).

Intense exercise can increase the number of circulating leukocytes (0.5–2.6-fold), monocytes (0.2–2.5-fold), granulocytes (0.3–4-fold) and neutrophils (0.8–2.6-fold) during and >1 h after exercise [3, 9–11]. Lymphocytes also increased by up to one-fold during exercise, but lymphocyte counts can be downregulated to below baseline concentrations (–19% –60%) for up to 6 h after an intense bout of exercise [3, 9, 10]. The increase in immune cell counts during exercise is due to the effects of hemodynamic shear stress, which demarginates (detaches) immune cells that have adhered to blood vessel walls, as well as from the lungs, liver and spleen, into systemic circulation [12]. Neutrophils are also mobilized from the bone marrow 1–3 h after exercise [11]. Elevated concentrations of exercise-induced catecholamines also bind to β_2 -adrenergic receptors on leukocytes which also contribute to their mobilization into systemic circulation [13]. The decrease in post-exercise lymphocyte count is accompanied by reduced concentrations in NK cells (–60% to –80%), B cells (–8% to –40%), T-cells (–14% to –50%), T-helper (–5% to –52%) and cytotoxic T (–40% to –60%) cells [10, 14–18]. The suppression in post-exercise immune cell count is due to the inter-compartmental shift of lymphocyte subtype cells, from central circulation to peripheral organs [4, 5], and is related to exercise intensity and volume, but not to the type of exercise [4, 5, 10, 14, 19–21]. The post-exercise mobilization of immune cells into peripheral organs such as the liver, lungs and skeletal muscle may be instrumental in initiating tissue remodelling, which can contribute to the metabolic and phenotypic changes in these organs.

Successful long-term adaptation to exercise training is predicated on the body's ability to mount and regulate an immune response appropriately during intense

exercise, including the modulation of the cytokine and chemokine responses. Failure of the immune system to properly regulate these responses, as a consequence of excessive training load or sub-optimal recovery, may suppress immune functions, compromise athletic performance and increase the risks of respiratory infections. This phenomenon is known as over-reaching when the effects are more transient, lasting days or weeks, and can progress to an over-trained state when the malaise lasts for months. In a previous study conducted in a small group ($N = 10$) of well-trained cyclists and triathletes [22], 28 days of continuous, intensified endurance training comprising primarily (~75–80%) of cycling, with the remaining time spent on running, swimming and weight lifting, did not result in changes in resting leukocyte cell counts across days 1, 14, 28 and 30. Differences in immune cell counts are actually observed between elite athletes and the reference adult populations when such outcomes are measured across a longer span of time. A cross-sectional study that tracked elite athletes from 14 sports (937 women, 1310 men) for 10 years demonstrated that well-trained and elite endurance athletes present with ~16% lower resting leukocyte and neutrophil counts compared to the general healthy population [23]. Of note, Horn's study detected clinical neutropenia (defined as $<2.0 \times 10^9/L$) in 17% and 16% of cyclists and triathletes, respectively, and 5% from the total athlete sample pool. It is unknown whether the athletes in the study experienced symptoms of over-training, as training diaries and medical visits were not documented. Nevertheless, exposures to a single bout of intense exercise and to an extended period of high training load (particularly endurance-type exercise) can induce immune suppression and increase the risks of infections. These research findings have important practical implications on the well-being of athletes who are exposed to prolonged high training loads. Sports and health professionals can contribute to the well-being of athletes by being aware of, and by taking precautions against the effects of prolonged intense training on the immune system.

This chapter aims to introduce the key concepts and research findings in exercise and immune health in active healthy populations. This topic has been reviewed in greater depth elsewhere, which readers can refer to for further reading [6, 13, 24–26]. Our primary focus is to summarize the current knowledge of exercise-induced immune response in endurance, strength and combined (concurrent) endurance and strength training. Specifically, we will describe exercise-induced cytokine responses, and how such inflammatory responses modulate infection risk and tissue homeostasis in healthy populations engaging in different modes of exercise.

Models of Exercise Immunology Research

“Open Window” and “J-Curve” Models of Exercise-Induced Immunosuppression

Interest in exercise immunology research has grown since the 1980s, with more than 3500 peer-reviewed publications currently in this domain of study [6]. Much of the bases for exercise immunology research in the last 30 years were influenced by the classical “open window” and the “J-curve” models. The “open window” model

suggests that immune functions can be suppressed for 3–72 h after an intense bout of exercise [27]. During this “open window” period, viruses that are dormant in the cells may be reactivated to cause an infection. Exposing the host to repeated bouts of exercise during this open window period would further suppress the immune system and increase the risk of having an infection [28, 29]. Scholars also subscribe to the concept of the “J” curve model (Fig. 9.1), suggesting that prolonged intense training is associated with a high risk of infection, followed by sedentary individuals with an average risk of infection, and lastly, individuals participating in moderate-intensity training having the lowest risk of infection [30, 31].

Both the “open window” and “J-curve” models were instrumental in the development of exercise immunology research, but recent evidence suggest that the concepts behind these classical models may be challenged.

Contrary to the classical models, recent data suggest that the risks of exercise-induced URTI can differ between athletes competing in international and national competitions [6]. The incidence of URTI, monitored over four training and competitive seasons (4 years), was 35–65% lower in international-level than in national-level swimmers [32]. In a mixed group of elite endurance athletes (cross-country skiers, biathletes, runners), data retrieved from 3 to 16 years of training logs showed that the number of self-reported sick days was inversely related to training volume [33]. Increasing training load by 20% for 2 weeks also did not alter immune response in endurance athletes before, during, and 2 h after running in warm and humid conditions [34]. These observations challenged the central doctrine proposed by the classical J-curve and open window models and led some scholars to suggest that besides lifestyle factors, the immune system in the higher performing athletes may be genetically selected to better tolerate exercise-induced stresses [6].

“Tissue Injury” Model of Exercise-Induced Immunosuppression

Another model to explain exercise-induced immune suppression is the “tissue injury” hypothesis [35]. This model proposes that tissue damage resulting from repetitive mechanical trauma during intense exercise training, combined with an

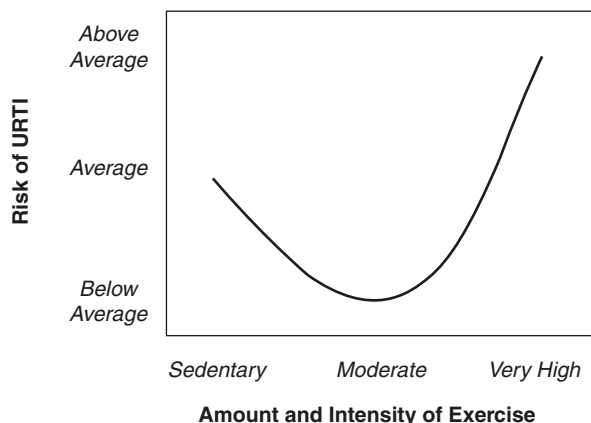


Fig. 9.1 “J-curve” model depicting the risks of developing URTI based on the volume and intensity of exercise training

accumulation of sub-optimal recovery can induce a pro-inflammatory response involving cytokines typically produced by T-helper (Th)1 cells, including interleukin (IL)-2, IL-6, interferon (IFN)- γ , and tumour necrosis factor (TNF)- α . The first wave of Th1 cytokine response is followed by a persistent anti-inflammatory cytokine response involving cytokines produced by Th2 cells, i.e. IL-4, IL-5, IL-10, and IL-13. The Th2 cytokine response is the host response to restore homeostasis and prevent further tissue damage by dampening the pro-inflammatory (Th1) cytokine response [36].

Adding to the concept of the tissue injury model is the current appreciation of the immunoregulatory role of endogenous stress molecules termed “alarmins”, which have been shown to respond to cellular insults, including oxidative stress, hypoxia, cellular necrosis, etc. [37]. Briefly, these alarmins, which include the prototypical danger protein, high mobility group box (HMGB) protein 1, are normally sequestered in either the nuclear or extracellular compartments of a cell. When cells experience stress or undergo necrosis, these alarmins are mobilized and released into the systemic circulation, where they behave like cytokines by binding to pattern recognition receptors (PRR)s on target cells [38]. Importantly, the release of alarmins can initiate signalling cascades for the ensuing cytokine and chemokine responses [39]. Few human studies have explored the role of alarmins in exercise-induced muscle damage but it is attractive to speculate that they may play a role in the immunoregulatory process in acute and long-term exercise.

Exercise-induced cytokine responses drive the recruitment of immune cells into the damaged muscle milieu, and neutrophils are the earliest immune cells to be recruited, followed by monocytes [40]. Gene and protein expression levels of exercise-induced chemokines, such as CX₃CL1, were upregulated after an acute bout of cycle ergometry [41], which in turn, may recruit circulating monocytes into the skeletal muscle microenvironment for remodelling and repair. These monocytes differentiate into skeletal muscle macrophages and cross-talk with myogenic precursor cells [42], which are essentially muscle-specific stem cells. Discrete macrophage populations within the skeletal muscle milieu, such as M1 and M2 cells, phagocytose damaged myofibres, secrete mitogenic factors, and participate in angiogenesis and matrix remodelling, which contributes to muscle cell regeneration [40].

The lingering presence of Th2 cytokines can also be a sign of unresolved chronic inflammation, which can suppress cell-mediated immune function against infections, including the downregulation of respiratory burst in neutrophils by IL-10 [43]. This Th2 response was observed during months of intense training in infection-prone high-performance athletes, who had fourfold higher circulating concentrations of the anti-inflammatory Th2 cytokines, particularly IL-10, compared with athletes free from infections [44]. In summary, the “tissue injury” model suggests that increased susceptibility to colds or allergies in athletes may be due to an imbalanced inflammatory response, partly due to the combination of sub-optimal recovery, high training load, and acute increase in training load or frequency. These imbalanced inflammatory responses can contribute to, and also be the consequences of the over-reaching/over-training phenomena in a positive feedback loop.

Immune Responses to Endurance Exercise (Acute and Chronic)

Acute Endurance Exercise

Acute endurance exercise is associated with leukocytosis [3, 9, 10] in healthy individuals, which is also modulated by their interactions with cytokines and chemokines. The magnitude of this cytokine response is influenced by the mode, duration, and intensity of exercise. Many acute exercise studies involving long duration and high-intensity exercises, such as marathons [45, 46], the Ironman triathlon [47], and long distance cycling (164 km road race; [48]) typically showed an increase in pro-(IL-1 β , IL-6, IL-8) and anti-inflammatory (IL-1ra, IL-10) cytokines.

Similar outcomes in cytokine response have been demonstrated with other exercise studies that are of lower intensities or shorter duration. For example, cycling 30 min to 3 h at 60–65% $\text{VO}_{2\text{ max}}$ [49], 80% $\text{VO}_{2\text{ peak}}$ [50], 90% of second ventilatory threshold [51] and 70% $\text{VO}_{2\text{ max}}$ [52] resulted in increased systemic circulating concentrations of IL-1 β , IL-1ra, IL-6, IL-8 IL-10, and TNF- α . These findings of enhanced pro- and anti-inflammatory cytokine responses are corroborated in other studies conducted in runners [53–55].

Transcriptional levels of inflammatory genes are also enhanced after endurance exercise in peripheral blood mononuclear cells (PBMCs) [56, 57] and neutrophils [54], suggesting that these immune cells may be the source of circulating cytokines and chemokines after exercise. In healthy men, running (80% $\text{VO}_{2\text{ max}}$) till exhaustion elicited an upregulation of 450 genes and down-regulation of 150 genes involved in stress (e.g. heat shock protein family A (HSPA1A)) and inflammatory responses (e.g. matrix metalloproteinase (MMP)-9) [56]. A less intense exercise bout comprising cycling at 80% $\text{VO}_{2\text{ peak}}$ for 30 min changed transcriptional output of 311 genes in PBMCs after exercise [57], wherein HSP70 was upregulated by >3.5-fold. Interestingly, the gene encoding the anti-inflammatory cytokine, IL-1ra, was upregulated by 1.5-fold during recovery, which paralleled the nearly 1.5-fold increase in circulating IL-1ra concentration at the same time point.

Despite the variations in temporal responses of individual pro-inflammatory and anti-inflammatory cytokines among different studies, the consensus is that acute endurance exercise stimulates an inflammatory response, which is necessary to mobilize leukocytes for tissue repair.

Chronic Endurance Exercise

The chronic effects of endurance exercise training have also been investigated in national-level athletes [58] and healthy individuals participating in 8 [59, 60] or 12 weeks [61] of endurance exercise training. In the young (18 years) athletic population, long-term participation (~ 8 year) in endurance exercise training was associated with a downregulation of pro-inflammatory cytokine gene expression (IL-8, IL-15) and chemokines (CXCL8, CXCL1) in PBMCs [58]. In healthy men and women training 4 days weekly for 8 weeks in aerobic activities (e.g. treadmill

running, cycling for 30–40 min) showed a decrease in inducible TNF- α response (24 h post-exercise), but only in participants stratified to the high-intensity (55–80% of HR_{max}) condition, compared with the moderate-intensity (55–60% of HR_{max}) condition [61]. In another study, running 3 times weekly for 8 weeks at 50–75% VO_{2 max} did not show changes in TNF- α concentrations immediately after exercise [60]. These discrepancies may be explained by the different exercise protocols, as well as time course for blood taking.

Immune Responses to Strength Training (Acute and Chronic)

Acute Strength Exercise

Whereas the effects of endurance training on inflammatory responses are well documented, fewer studies on strength training and inflammatory paradigms have been conducted in healthy populations. In general, the leukocyte response to acute resistance exercise is similar to that in endurance exercise. For example, a single bout of resistance exercise (10 repetitions of squats at 65% 1-repetition maximum (RM)) performed by healthy middle-aged men (46.9 ± 1.2 years) until volitional fatigue enhanced leukocytosis [17]. Acute enhancement in leukocytosis was also observed in other resistance exercise studies with younger volunteers [62–64]. These studies also demonstrated increased systemic concentrations of IL-6 [62], TNF- α [62, 64], and chemokines, e.g. monocyte chemotactic protein (MCP)-1 [64]. The increased concentration of MCP-1 in the systemic circulation after resistance exercise supports the recruitment of monocytes [64] into skeletal muscle, such as CD14⁺⁺CD16⁻ monocytes, which are the primary immune cells that secrete IL-10 [65].

The extent of total muscle fibre recruitment *via* the manipulation of the number of sets or repetitions to shift the emphasis between a hypertrophic and a maximal all-out effort contraction can result in differences in leukocytosis and cytokine responses after an acute bout of resistance exercise [63]. Increased leukocytosis was observed immediately after acute hypertrophic bout of exercise (leg press with 5 sets of 10 reps at 80% of 1RM). Comparatively, an increased leukocyte count was only observed after 30 min after the maximal bout of leg press (15 sets of 1 rep at 100% of 1RM). The chemokine, MCP-1, and anti-inflammatory cytokine, IL-1ra, decreased and increased, respectively, after the hypertrophic bout, but no changes were observed in these inflammatory markers after the maximal bout [63]. This intriguing study suggests that the recruitment of immune cells into the circulation, possibly for skeletal muscle remodelling may occur differently, depending on the degree of muscle fibre recruitment, and whether the contractions are brief or sustained.

Chronic Strength Exercise

The inflammatory response to short-term resistance exercise training varies, depending on the frequency, intensity, and duration of training. Three sessions of resistance

training weekly at moderate- (45–55% of 1RM) or high-intensity (80–90% of 1RM) over 6 weeks had no effect on systemic IL-6 and IL-1 β concentrations in young healthy men [66]. However, moderate-intensity training (60% of 1RM) performed for 8 weeks by college-aged males resulted in higher systemic IL-8 concentrations than in either the control or high-intensity training groups [67]. High-intensity resistance training (80% of 1RM) also resulted in higher concentrations of serum soluble TNF receptor (sTNFR1), IL-1ra and IL-8 in healthy men and women (36 $\pm$ 2 years) [68]. However, the low-intensity training (range of 20–40% of 1RM) showed decreased concentration of serum IL-6 after 9 weeks of training [68]. In contrast, 12 weeks of high-intensity resistance exercise (80% of 1RM; 2 days/week) had no effect on systemic concentration of IL-1 β , IL-2, IL-6, and TNF- α in healthy young men and women [69].

In a different study conducted in young healthy men [70], the authors investigated the effects of 12 weeks of resistance training on systemic cytokine and chemokine responses to two acute bouts of leg press exercise: either a hypertrophic (5 sets $\times$ 10 repetitions at 60% of 1RM) or power-type (10 sets of 5 repetitions at 60% of 1RM) exercise tests were completed by the participants. In the untrained state, serum IL-1 β , plasma IL-6, and resistin all increased immediately after either type of exercise. After 12 weeks of resistance training however, only plasma IL-1ra was increased after both exercise protocols, while plasma MCP-1 was increased only at 24 and 48 h.

These seemingly disparate findings on resistance training and inflammation may be due to different subject characteristics and exercise regimes. The extent of eccentric and concentric muscle action in resistance training may determine the remodelling process in skeletal muscle. Eccentric muscle action, but not concentric action resulted in increased MCP-1 and skeletal muscle stem cells [71] within skeletal muscle of young men. Hence, it is possible that resistance exercises that employ more concentric actions may elicit a different pattern of inflammatory cytokine and chemokine production, which may not result in extensive skeletal muscle damage and remodelling.

Immune Responses to Concurrent Strength and Endurance Exercise

Compared with research into the immune responses to endurance or strength exercise alone, considerably fewer studies have been conducted to determine the immune response to concurrent strength and endurance exercise. Therefore, a critical gap exists in the knowledge-base on how healthy individuals respond to combined strength and endurance exercises, given that many recreational and elite athletes perform some kind of combined training.

Acute Concurrent Exercise

Current research evidence suggests that combining both endurance and strength exercises acutely upregulates IL-1 β , IL-6, TNF- α , and TNF-R1 gene expression

in skeletal muscle, and these gene expression levels return to resting levels 48 h after the exercise session [72]. These responses differ from performing endurance or resistance exercises independently, where the gene expression for the same set of cytokines remained elevated [72]. In recreational weight-lifters, concurrent exercise performed in either order (strength-endurance or endurance-strength) resulted in increased systemic concentration of IL-6, but concentration of TNF- α was increased only in the endurance-strength condition [73].

Chronic Concurrent Exercise

Long-term concurrent training appears to have varied immune responses, depending on the structure of the programme. For middle-aged men, 12 weeks of concurrent training resulted in lower plasma concentrations of TNF- α and IL-6 [72], but 16 weeks of concurrent training showed no changes in resting plasma concentrations of these two cytokines [74, 75]. In another training study that spanned 24 weeks, healthy young men were stratified into two interventional and one control groups [76] where one group of subjects performed concurrent endurance and resistance training in the same session, 2–3 times per week, and the second intervention group performed alternating days of endurance and resistance training, 4–5 times per week. Performing concurrent endurance and resistance exercises in the same sessions or on alternate days resulted in a decrease in plasma leptin, resistin, and C-reactive protein (CRP). However, only subjects performing concurrent training on alternate days presented with decreased plasma IL-6, MCP-1, and TNF- α .

It is noteworthy that many authors have investigated concurrent endurance and resistance exercises using different modes of exercises, which makes the interpretation of the results difficult to generalize. In the study by Donges et al. [72], the authors implemented cycle exercise (55% of peak aerobic workload) as the endurance component, and leg extension exercise (8 sets of 8 repetitions at 70% maximal strength) as the strength component, whereas Libardi and colleagues [74, 75] used walking and running (55–85% of $\text{VO}_{2\text{ peak}}$) for endurance training combined with whole-body resistance exercises (3 sets of 8–10 repetitions). Finally, Ihalainen and colleagues [70, 76] employed a combination of cycle exercise (performed below and above anaerobic threshold) and whole-body resistance training with an emphasis on lower limb strength training.

The aforementioned studies illustrate a large diversity in study design, subject demographics, and immune outcomes regarding the science of concurrent training. Notwithstanding these disparate findings, some observations are still worth noting. For instance, it would appear that the amount of muscle mass recruited during training may be a factor that affects the immune responses to concurrent exercise training. In addition, the manipulation of rest and implementation of concurrent training sessions either within the same day or on alternate days may affect the dynamics of cytokine and chemokine response, as seen from the reduced MCP-1, IL-6, and TNF- α responses with alternating days of training, but not with same-day sessions [76].

This also suggests that the extent of leukocyte recruitment for skeletal muscle remodelling may be affected by such manipulations as well.

Finally, compared with younger men (18–35 years), older men (65–85 years) demonstrated an accentuated TNF- α response both at baseline and after 12 weeks of concurrent training [77], suggesting that in addition to exercise frequency, intensity, duration, and order (endurance-strength or strength-endurance), the age of individuals participating in concurrent exercise will influence their immune responses.

Summary

In this chapter, we have covered the general scope of exercise immunology and discussed the three models of immune responses to exercise that shape the prevailing research trends in our field. We have also presented immune responses to endurance, resistance, and concurrent exercise training, both in terms of acute exposures and short-term periods. The implications of immune responses on clinical or performance outcomes after resistance and concurrent exercise training are less clear and warrant further research.

In summary, a translational implication of exercise-induced inflammation in healthy individuals and athletes, is that intensified endurance training can lead to over-reaching or over-training symptoms, which increase the risk of developing acute URTIs. As such, the pro- and anti-inflammatory cytokine balance might determine the host defence to infection as well as tissue damage following exercise and training. Further research is necessary in the future and may entail the understanding of cross-talk between immune cells and their putative target organs/tissues.

References

1. Janeway CA, Travers P, Walport M, Shlomchik M. Immune biology. New York: Garland Publishing; 2001. p. 12, 69–87, 360–375, 683.
2. Mackinnon LT. Advances in exercise immunology, vol. 314. Champaign IL: Human Kinetics; 1999. p. 33.
3. Lim CL, Byrne C, Chew SA, Mackinnon LT. Leukocyte subset responses during exercise under heat stress with carbohydrate or water intake. *Aviat Space Environ Med.* 2005;76:726–32.
4. Nieman DC. Exercise effects on systemic immunity. *Immunol Cell Biol.* 2000a;78:496–501.
5. Nieman DC. Is infection risk linked to exercise workload? *Med Sci Sports Exerc.* 2000b;32(Suppl):S406–11.
6. Walsh NP, Samuel JO. Exercise, immune function and respiratory infection: an update on the influence of training and environmental stress. *Immunol Cell Biol.* 2016;94:132–9.
7. Suzuki K, Nakaji S, Yamada M, Totsuka M, Sato K, Sugawara K. Systemic inflammatory response to exhaustive exercise. *Cytokine kinetics. Exerc Immunol Rev.* 2002;8:6–48.
8. Lancaster GI, Jentjens RL, Moseley L, Jeukendrup AE, Gleeson M. Effect of pre-exercise carbohydrate ingestion on plasma cytokine, stress hormone, and neutrophil degranulation responses to continuous, high-intensity exercise. *Int J Sport Nutr Exerc Metab.* 2003;13:436–53.
9. Nieman DC, Buckley KS, Henson DA, Warren BJ, Suttles J, Ahle JC, Nehlsen-Cannarella SL. Immune function in marathon runners versus sedentary controls. *Med Sci Sports Exerc.* 1995a;27:986–92.

10. Nieman DC, Nehlsen-Cannarella SL, Donohue KM, Chritton DBW, Haddock BL, Stout RW, Lee JW. The effects of acute moderate exercise on leukocyte and lymphocyte subpopulations. *Med Sci Sports Exerc.* 1991;23:578–85.
11. Suzuki K, Totsuka M, Nakaji S, Yamada M, Kudoh S, Liu Q, Sugawara K, Yamaya K, Sato K. Endurance exercise causes interaction among stress hormones, cytokines, neutrophil dynamics, and muscle damage. *J Appl Physiol.* 1999;87:1360–7.
12. Simpson RJ, Kunz H, Agha N, Graff R. Exercise and the regulation of immune functions. *Prog Mol Biol Transl Sci.* 2015;135:355–80.
13. Peake J, Neubauer O, Walsh NP, Simpson RJ. Recovery of the immune system after exercise. *J Appl Physiol.* 2017;122:1077–87.
14. Gabriel H, Schwarz L, Steffens G, Kindermann W. Immunoregulatory hormones, circulating leukocyte and lymphocyte subpopulations before and after endurance exercise of different intensities. *Int J Sports Med.* 1992;13:359–66.
15. Henson DA, Nieman DC, Parker JCD, Rainwater MK, Butterworth DE, Warren BJ, Utter AC, Davies JM, Fagoaga OR, Nehlsen-Cannarella SL. Carbohydrate supplementation and lymphocyte proliferative response to long endurance running. *Int J Sports Med.* 1998;19:574–80.
16. Nehlsen-Cannarella SL, Nieman DC, Balk-Lamberton AJ, Markoff PA, Chritton DBW, Gusewitch G, Lee JW. The effects of moderate exercise training on immune response. *Med Sci Sports Exerc.* 1991;23:64–70.
17. Nieman DC, Henson DA, Sampson CS, Herring JL, Suttles J, Conley M, Stone MH, Butterworth DE, JM D. The acute immune response to exhaustive resistance exercise. *Int J Sports Med.* 1995b;16:322–8.
18. Nieman DC, Simandle S, Henson DA, Warren BJ, Suttles J, Davis JM, Buckley KS, Ahle JC, Butterworth DE, Fagoaga OR, Nehlsen-Cannarella SL. Lymphocyte proliferative response to 2.5 hours of running. *Int J Sports Med.* 1995c;16:404–8.
19. Kargotich S, Keast D, Goodman C, Crawford GPM, Morton AR. The influence of blood volume changes on leukocyte and lymphocyte subpopulations in elite swimmers following interval training of varying intensities. *Int J Sports Med.* 1997;18:373–80.
20. Nieman DC. Immune response to heavy exertion. *J Appl Physiol.* 1997;82:1385–94.
21. Tvede N, Kappel M, Halkjoer-Kristensen J, Galbo H, Pedersen BK. The effects of light, moderate, and severe bicycle exercise on lymphocyte subsets, natural and lymphokine killer cells, lymphocyte proliferative response and interleukin 2 production. *Int J Sports Med.* 1993;15:275–82.
22. Ndon JA, Snyder AC, Foster C, Wehrenberg WB. Effects of chronic intense exercisettraining on the leukocyte response to acute exercise. *Int J Sports Med.* 1992;13:176–82.
23. Horn PL, Pyne DB, Hopkins WG, Barnes CJ. Lower white blood cell counts in elite athletes training for highly aerobic sports. *Eur J Appl Physiol.* 2010;110:925–32.
24. Gleeson M, Pyne DB. Respiratory inflammation and infection in high performance athletes. *Immunol Cell Biol.* 2016;94:124–31.
25. Walsh NP, Gleeson M, Pyne DB, Nieman D, Dhabhar FS, Shephard RJ, Oliver SJ, Bermon S, Kajeniene A. Position statement part II: maintaining immune health. *Exerc Immunol Rev.* 2011a;17:64–103.
26. Walsh NP, Gleeson M, Shephard RJ, Gleeson M, Wood JA, Bishop NC, Fleshner M, Green C, Pedersen BK, Hoffman-Goetz L, Rogers CJ, Northoff GH, Abbassi A, Simon P. Position statement part I: immune function and exercise. *Exerc Immunol Rev.* 2011b;17:6–63.
27. Pedersen BK, Rohde T, Zacho M. Immunity in athletes. *J Sports Med Phys Fitness.* 1996;36:236–45.
28. Mackinnon LT. Overtraining effects on immunity and performance in athletes. *Immunol Cell Biol.* 2000;78:502–9.
29. Smith LL. Overtraining, excessive exercise, and altered immunity. *Sports Med.* 2003;33:347–64.
30. Nieman DC. Exercise, immunology and nutrition. *World Rev Nutr Diet.* 2001;90:89–101.
31. Nieman DC, Nehlsen-Cannarella SL. The immune response to exercise. *Semin Hematol.* 1994;31:166–79.

32. Hellard P, Avalos M, Guimaraes F, Toussaint JF. Training-related risk of common illness in elite swimmers over a 4-year period. *Med Sci Sports Exerc.* 2015;47:698–707.
33. Martensson S, Nordebo K, Malm C. High training volumes are associated with a low number of self-reported sick days in elite endurance athletes. *J Sports Sci Med.* 2014;13:929–33.
34. Lim CL, Pyne DB, Horn P, Kalz A, Saunders P, Peake J, Suzuki K, Wilson G, Mackinnon LT. The effects of increased endurance training load on biomarkers of heat tolerance during intense exercise in the heat. *Appl Physiol Nutr Metab.* 2009;34:616–24.
35. Smith LL. Tissue trauma: the underlying cause of overtraining syndrome? *J Strength Cond Res.* 2004;18(1):185–93.
36. Suzuki K, Nakaji S, Kurakake S, Totsuka M, Sato K, Kuriyama T, Fujimoto H, Shibusawa K, Machida K, Sugawara K. Exhaustive exercise and type-1/type-2 cytokine balance in special focus on interleukin-12 p40/p70. *Exerc Immunol Rev.* 2003a;9:48–57.
37. Oppenheim JJ, Yang D. Alarmins: chemotactic activators of immune responses. *Curr Opin Immunol.* 2005;17:359–65.
38. Hung YL, Suzuki K. The pattern recognition receptors and lipopolysaccharides (LPS)-induced systemic inflammation. *Int J Res Stud Med Health Sci.* 2017;7:1–7.
39. Yang D, Han Z, Oppenheim JJ. Alarmins and immunity. *Immunol Rev.* 2017;280:41–56.
40. Juban G, Chazaud B. Metabolic regulation of macrophages during tissue repair: insights from skeletal muscle regeneration. *FEBS Lett.* 2017. doi: <https://doi.org/10.1002/1873-3468.12703>.
41. Strömberg A, Rullman E, Jansson E, Gustafsson T. Exercise-induced upregulation of endothelial adhesion molecules in human skeletal muscle and number of circulating cells with remodeling properties. *J Appl Physiol.* 2017;122:1145–54.
42. Saclier M, Yacoub-Youssef H, Mackey AL, Arnold L, Ardjoune H, Magnan M, Sailhan F, Chelly J, Pavlath GK, Mounier R, Kjaer M, Chazaud B. Differentially activated macrophages orchestrate myogenic precursor cell fate during human skeletal muscle regeneration. *Stem Cells.* 2013;31:384–96.
43. Dang PM, Elbim C, Marie JC, Chiandotto M, Gougerot-Pocidallo MA, El-Benna J. Anti-inflammatory effect of interleukin-10 on human neutrophil respiratory burst involves inhibition of GM-CSF-induced p47PHOX phosphorylation through a decrease in ERK1/2 activity. *FASEB J.* 2006;20:1504–6.
44. Gleeson M, Bishop N, Oliveira M, McCauley T, Tauler P, Muhamad AS. Respiratory infection risk in athletes: association with antigen-stimulated IL-10 production and salivary IgA secretion. *Scand J Med Sci Sports.* 2012;22:410–7.
45. Suzuki K, Nakaji S, Yamada M, Liu Q, Kurakake S, Okamura N, Kumae T, Umeda T, Sugawara K. Impact of a competitive marathon race on systemic cytokine and neutrophil responses. *Med Sci Sports Exerc.* 2003b;35:348–55.
46. Suzuki K, Yamada M, Kurakake S, Okamura N, Yamaya K, Liu Q, Kudoh S, Kowatari K, Nakaji S, Sugawara K. Circulating cytokines and hormones with immunosuppressive but neutrophil-priming potentials rise after endurance exercise in humans. *Eur J Appl Physiol.* 2000;81:281–7.
47. Suzuki K, Peake J, Nosaka K, Okutsu M, Abbiss CR, Surriano R, Bishop D, Quod MJ, Lee H, Martin DT, Laursen PB. Changes in markers of muscle damage, inflammation and HSP70 after an Ironman Triathlon race. *Eur J Appl Physiol.* 2006;98:525–34.
48. Luk HY, Levitt DE, Lee EC, Ganio MS, McDermott BP, Kupchak BR, McFarlin BK, Hill DW, Armstrong LE, Vingren JL. Pro- and anti-inflammatory cytokine responses to a 164-km road cycle ride in a hot environment. *Eur J Appl Physiol.* 2016;116:2007–15.
49. Moldoveanu AI, Shephard RJ, Shek PN. Exercise elevates plasma levels but not gene expression of IL-1beta, IL-6, and TNF-alpha in blood mononuclear cells. *J Appl Physiol.* 2000;89:1499–504.
50. Zaldivar F, Wang-Rodriguez J, Nemet D, Schwindt C, Galassetti P, Mills PJ, Wilson LD, Cooper DM. Constitutive pro- and anti-inflammatory cytokine and growth factor response to exercise in leukocytes. *J Appl Physiol.* 2006;100:1124–33.
51. Kakanis MW, Peake J, Brenu EW, Simmonds M, Gray B, Marshall-Gradisnik SM. T helper cell cytokine profiles after endurance exercise. *J Interf Cytokine Res.* 2014;34:699–706.

52. Ulven SM, Foss SS, Skjølsvik AM, Stadheim HK, Myhrstad MC, Raael E, Sandvik M, Narverud I, Andersen LF, Jensen J, Holven KB. An acute bout of exercise modulate the inflammatory response in peripheral blood mononuclear cells in healthy young men. *Arch Physiol Biochem*. 2015;121:41–9.
53. Landers-Ramos RQ, Jenkins NT, Spangenburg EE, Hagberg JM, Prior SJ. Circulating angiogenic and inflammatory cytokine responses to acute aerobic exercise in trained and sedentary young men. *Eur J Appl Physiol*. 2014;114:1377–84.
54. Neubauer O, Sabapathy S, Lazarus R, Jowett JB, Desbrow B, Peake JM, Cameron-Smith D, Haseler LJ, Wagner KH, Bulmer AC. Transcriptome analysis of neutrophils after endurance exercise reveals novel signaling mechanisms in the immune response to physiological stress. *J Appl Physiol*. 2013;114:1677–88.
55. Sugama K, Suzuki K, Yoshitani K, Shiraishi K, Miura S, Yoshioka H, Mori Y, Kometani T. Changes of thioredoxin, oxidative stress markers, inflammation and muscle/renal damage following intensive endurance exercise. *Exerc Immunol Rev*. 2015;21:130–42.
56. Buttner P, Mosig S, Lechtermann A, Funke H, Mooren FC. Exercise affects the gene expression profiles of human white blood cells. *J Appl Physiol*. 2007;102:26–36.
57. Connolly PH, Caiozzo VJ, Zaldivar F, Nemet D, Larson J, Hung SP, Heck JD, Hatfield GW, Cooper DM. Effects of exercise on gene expression in human peripheral blood mononuclear cells. *J Appl Physiol*. 2004;97:1461–9.
58. Liu D, Wang R, Grant AR, Zhang J, Gordon PM, Wei Y, Chen P. Immune adaptation to chronic intense exercise training: new microarray evidence. *BMC Genomics*. 2017;18(1):29. <https://doi.org/10.1186/s12864-016-3388-5>.
59. Elmer DJ, Laird RH, Barberio MD, Pascoe DD. Inflammatory, lipid, and body composition responses to interval training or moderate aerobic training. *Eur J Appl Physiol*. 2016;116:601–9.
60. Jahromi AS, Zar A, Ahmadi F, Krstrup P, Ebrahim K, Hovanloo F, Amani D. Effects of endurance training on the serum levels of tumour necrosis factor- α and interferon- γ in sedentary men. *Immune Netw*. 2014;14:255–9.
61. Sloan RP, Shapiro PA, Demeersman RE, McKinley PS, Tracey KJ, Slavov I, Fang Y, Flood PD. Aerobic exercise attenuates inducible TNF production in humans. *J Appl Physiol*. 2007;103:1007–11.
62. Brunelli DT, Caram K, Nogueira FR, Libardi CA, Prestes J, Cavaglieri CR. Immune responses to an upper body tri-set resistance training session. *Clin Physiol Funct Imaging*. 2014;34:64–71.
63. Ihalainen J, Walker S, Paulsen G, Häkkinen K, Kraemer WJ, Härmäläinen M, Vuolteenaho K, Moilanen E, Mero AA. Acute leukocyte, cytokine and adipocytokine responses to maximal and hypertrophic resistance exercise bouts. *Eur J Appl Physiol*. 2014;114:2607–16.
64. Wells AJ, Hoffman JR, Jajtner AR, Varanoske AN, Church DD, Gonzalez AM, Townsend JR, Boone CH, Baker KM, Beyer KS, Mangine GT, Oliveira LP, Fukuda DH, Stout JR. Monocyte recruitment after high-intensity and high-volume resistance exercise. *Med Sci Sports Exerc*. 2016;48:1169–78.
65. Ziegler-Heitbrock HW, Ströbel M, Kieper D, Fingerle G, Schlunck T, Petersmann I, Ellwart J, Blumenstein M, Haas JG. Differential expression of cytokines in human blood monocyte subpopulations. *Blood*. 1992;79:503–11.
66. Sheikholeslami Vatani D, Ahmadi S, Ahmadi Dehrashid K, Gharibi F. Changes in cardiovascular risk factors and inflammatory markers of young, healthy, men after six weeks of moderate or high intensity resistance training. *J Sports Med Phys Fitness*. 2011;51:695–700.
67. Yeo NH, Woo J, Shin KO, Park JY, Kang S. The effects of different exercise intensity on myokine and angiogenesis factors. *J Sports Med Phys Fitness*. 2012;52:448–54.
68. Forti LN, Van Roie E, Njemini R, Coudyzer W, Beyer I, Delecluse C, Bautmans I. Effects of resistance training at different loads on inflammatory markers in young adults. *Eur J Appl Physiol*. 2017;117:511–9.
69. Rall LC, Roubenoff R, Cannon JG, Abad LW, Dinarello CA, Meydani SN. Effects of progressive resistance training on immune response in aging and chronic inflammation. *Med Sci Sports Exerc*. 1996;28:1356–65.

70. Ihalainen J, Ahtiainen JP, Walker S, Paulsen G, Selänne H, Härmäläinen M, Moilanen E, Peltonen H, Mero AA. Resistance training status modifies inflammatory response to explosive and hypertrophic resistance exercise bouts. *J Physiol Biochem.* 2017a;73:595–604.
71. Hyldahl RD, Olson T, Welling T, Groscost L, Parcell AC. Satellite cell activity is differentially affected by contraction mode in human muscle following a work-matched bout of exercise. *Front Physiol.* 2014;5(485) <https://doi.org/10.3389/fphys.2014.00485>.
72. Donges CE, Duffield R, Smith GC, Short MJ, Edge JA. Cytokine mRNA expression responses to resistance, aerobic, and concurrent exercise in sedentary middle-aged men. *Appl Physiol Nutr Metab.* 2014;39:130–7.
73. Inoue DS, Panissa VL, Monteiro PA, Gerosa-Neto J, Rossi FE, Antunes BM, Franchini E, Cholewa JM, Gobbo LA, Lira FS. Immunometabolic responses to concurrent training: the effects of exercise order in recreational weightlifters. *J Strength Cond Res.* 2016;30:1960–7.
74. Libardi CA, Souza GV, Gáspari AF, Dos Santos CF, Leite ST, Dias R, Frollini AB, Brunelli DT, Cavaglieri CR, Madruga VA, Chacon-Mikahil MP. Effects of concurrent training on interleukin-6, tumour necrosis factor-alpha and C-reactive protein in middle-aged men. *J Sports Sci.* 2011;29:1573–81.
75. Libardi CA, De Souza GV, Cavaglieri CR, Madruga VA, Chacon-Mikahil MP. Effect of resistance, endurance, and concurrent training on TNF- α , IL-6, and CRP. *Med Sci Sports Exerc.* 2012;44:50–6.
76. Ihalainen JK, Schumann M, Eklund D, Härmäläinen M, Moilanen E, Paulsen G, Häkkinen K, Mero AA. Combined aerobic and resistance training decreases inflammation markers in healthy men. *Scand J Med Sci Sports.* 2017b; <https://doi.org/10.1111/sms.12906>.
77. Stewart LK, Flynn MG, Campbell WW, Craig BA, Robinson JP, Timmerman KL, McFarlin BK, Coen PM, Talbert E. The influence of exercise training on inflammatory cytokines and C-reactive protein. *Med Sci Sports Exerc.* 2007;39:1714–9.